Microeconomics | **Professor Lars Stole**
Business 33801 | **Autumn 2024**

Collection of problems for Lectures 6-9

Below are a collection of 57 problems related to the material Lectures 6 through 9. At the end of Lectures 6, 7 and 8, I will assign a few questions from this set as the group assignment. Answers to the each assignment are due before the start of the next class. We will post solutions for all of these problems as the week progresses. There will be no assignment for the material in Lecture 9, but please study the questions and suggested solutions when preparing for the final exam.

These are group assignments. Each group should submit one solution set for each assignment. The name of every group member who participated in the exercise should be printed at the top of the assignment. One person from the group must attest that the names of the participating members are accurate. If possible, that person should sign the document. You may hand write (rather than type) your solutions – whichever is more convenient. Then scan/photograph your solutions and upload them to the canvas course website.

Problem 1 Consider the taxicab example from class. Suppose that the market demand increases to

$$P(Q) = 1.2 - \frac{Q}{100,000},$$

but everything else remains unchanged. That is, there are 6 non-transferable taxi licenses and each taxi driver has the same opportunity cost function given by $C(q) = 0.20q$ with a capacity of 10,000 miles. What is (a) the equilibrium price, (b) the economic profit each firm earns from owning the license? (c) the deadweight loss of restricting the number of licenses to 6?

Problem 2 In the competitive market for DVDs, market demand is given by $Q^d = 500 - 20p$ and quantity supplied is given by $Q^s = 5p$, where quantity is given in millions of DVDs sold per year.

(a). What is the market equilibrium price and quantity?

(b). Suppose the government decided to tax DVD sellers by an amount of \$5 per DVD sold. (1) Under the tax, what after-tax price do consumers pay? (2). What after-tax price do sellers receive? (3). What is the new equilibrium quantity?

(c). Describe, in words or with graphs, how does this tax generate a deadweight loss? That is, what is the source or story behind the deadweight loss?

Problem 3 True, False, Uncertain? [Explain your answer.] In a paper by University of Chicago Professor Sam Peltzman, it is shown that for supermarkets, approximately 40% of any reduction in marginal costs is passed along to consumers through a reduction in prices. *Therefore, supply is relatively more elastic than demand.*

Problem 4 True, False, Uncertain? [Explain your answer.] The State of Illinois removes a 5 cent per gallon tax on gasoline (legally paid by the gasoline suppliers) because gasoline prices reach record highs. The demand elasticity of gasoline is -0.5. The supply elasticity of gasoline is 1.5. *The majority of the tax reduction to gasoline suppliers will be passed onto consumers.*

Problem 5 Assume that the short-run market demand curve for cartons of cigarettes is given by the equation:

$$Q^d = 100,000,$$

while the supply curve of cartons of cigarettes is given by the equation:

$$Q^s = 20,000 + 20,000p.$$

Assume that the government imposes a tax of \$1 per carton on the purchase of cigarettes.

(a). What is the equilibrium price and quantity before the tax?

(b). What is the <u>total</u> transaction price, p_s, received by the sellers, what is the <u>total</u> transaction price, p_b, paid by the buyers, and what is the after-tax quantity transacted?

(c). What is the excess burden (deadweight loss) of this tax? Explain economically.

(d). Assume that the demand of cigarettes is more elastic in the long run than in the short run. (i) How does the size of excess burden change from the short run to the long run? (ii) How does the distribution of the tax burden change from the short run to the long run? Briefly explain each.

Problem 6 Assume that the market demand curve for honeydew melons is given by the equation:

$$Q^d = 100,000 - 40,000p,$$

while the short-run supply curve of these melons is given by the equation:

$$Q^s = 20,000.$$

Assume that the government imposes a tax of \$1 per melon on the purchase of melons.

(a). What are the equilibrium price and quantity before the tax is imposed?

(b). What are the elasticities of supply and demand at the pre-tax equilibrium price and quantity from (a) above?

(c). What are the equilibrium price and quantity after the tax is imposed?

(d). How much tax revenue is raised by the government?

(e). How much of the economic incidence of the tax is borne by consumers and how much by producers?

(f). What is the deadweight loss of this tax? Explain.

(g). Assume that the supply of melons is more elastic in the long run than in the short run. How does the size of the deadweight loss change from the short run to the long run? How does the distribution of the economic tax incidence change from the short run to the long run? Explain.

Problem 7 True, False, or Uncertain. [Explain your answer!] It has recently been argued that consumers see very little of gasoline taxes passed along into the price. Assume that the gasoline market is perfectly competitive and the taxes are levied per gallon. *This result implies that the supply of gasoline must be very elastic.*

Problem 8 True, False, or Uncertain. [Explain your answer!] If the price of steel sold by the steel mills is held below the market equilibrium by a government regulation (<u>but</u> the resale of steel is unregulated by the government), *then price of automobiles will rise.*

Problem 9 Suppose that the demand and supply for coffee are given by equations

$$Q^d = 1200 - 200p$$

$$Q^s = 200p.$$

(a). Find the equilibrium price and quantity for this market.

(b). Suppose that the government decides to buy up as much coffee as it takes to raise the market price to $p = 4$. Furthermore, suppose the government destroys all of its purchased coffee. How much deadweight loss does this intervention generate?

(c). Suppose instead that the government decides to subsidize coffee production by paying producers \$1/bushel for every bushel of coffee sold. What are the after-subsidy equilibrium prices that the buyers effectively pay, p_b^*, and that the sellers effectively receive, p_s^*?

Problem 10 True, False, or Uncertain. [Explain your answer!] Consider a constant-cost industry with homogeneous firms in long-run equilibrium, with price equal to $p = 1$. Suppose that the government places a price ceiling on output at $p = \frac{1}{2}$, and that this law is effective at preventing trade from occurring at higher prices than $p = \frac{1}{2}$. *The long-run equilibrium output of this market is unchanged by the price restriction.*

Problem 11 True, False, Uncertain? [Explain your answer.] Suppose that demand for olives is elastic and the government decides to encourage production by offering olive growers a subsidy for each kilo of olives sold at market. *Total market expenditures by consumers on olives decrease following the subsidy.*

Problem 12 Consider a perfectly competitive labor market with downward sloping labor demand (from firms) and upward sloping labor supply (from workers). Suppose that the government institutes a minimum wage law which requires firms to pay $\underline{w}$ per hour which is more than the current market equilibrium of w^* per hour. Indicate in a graph (or graphs) how the market outcome changes (e.g., wages, labor transacted, etc.) and indicate how surpluses change for all market participants (workers and firms, in and out of the market). Also illustrate the deadweight loss from the wage law and indicate (verbally) why it emerges?

Problem 13 Which of the following statements is *NOT* generally true regarding a binding price ceiling in a competitive market? Explain your answer. [No credit will be given without a suitable explanation.]

(a). All producer surplus is transferred to consumers.

(b). The market will not clear due to excess demand of the good.

(c). Producer surplus decreases when compared to the market before the price ceiling.

(d). Consumer surplus may increase or decrease.

Problem 14 True, False, Uncertain? [Explain your answer.] *If a government wishes to minimize the deadweight loss of raising a fixed amount of tax revenues, it should focus its taxes on goods whose demand or supply are relatively inelastic.*

Problem 15 True, False, or Uncertain. [Explain your answer!] *In a perfectly competitive, constant-cost industry, the imposition of a tax of \$2/unit will cause the buyer's total transaction price of the good (price paid to seller plus tax paid to government) to rise by \$2 in the long run.*

Problem 16 Suppose that a monopolist's *opportunity* costs are given by the function $C(q) = 16 + 4q$ and the monopolist's demand is given by $q^d = 40 - 2p$. What is the firm's optimal choice of p and q?; what are the corresponding profits?

Problem 17 True, False or Uncertain? [Explain your answer.] A monopolist with constant marginal cost currently sells to a market with demand function given by $q^d = D(p)$. *If the monopolist's market size doubles so that demand becomes $2D(p)$ (i.e., twice as much is demanded as before for any given price), then monopolist's optimal price will also double.* <u>Hint:</u> Consider the case of linear demand, $D(p) = a - bp$, to get intuition.

Problem 18 True, False, or Uncertain? [Explain your answer!] *In the long run, a monopolist will choose to produce at the point where long-run average cost is minimized.*

Problem 19 The local monopoly coffee and espresso shop sells regular coffee (denoted with a "*c*") and premium espresso (denoted with an "*e*"). The market demand functions for coffee and espresso are, respectively,

$$q_c^d = 12 - 3p_c + p_e,$$
$$q_e^d = 12 + p_c - 3p_e,$$

where quantities are measured in cups per week and prices are measured in dollars per cup. Note, we can also write this demand system as a pair of demand "curves" with prices on the lefthand side:

$$p_c = 6 - \frac{3}{8}q_c - \frac{1}{8}q_e,$$
$$p_e = 6 - \frac{1}{8}q_c - \frac{3}{8}q_e.$$

(a). Are these products substitutes or complements? Explain.

(b). Suppose that the marginal and average costs of producing both products are constant. Regular coffee costs \$0.50 per cup to produce, and espresso costs \$1.50 per cup to produce. What are the profit-maximizing prices of regular coffee and espresso? (Hint: Write the monopolist's profit function as either a function of quantities (q_c and q_e) and maximize with respect to these, or write the monopolist's profit as a function of prices (p_c and p_e) and maximize with respect to these. The second approach is more direct for this question where we are asked for the optimal price and the marginal costs are constant.)

Problem 20 A monopolist's demand curve is given by $p = 1000 - q$. Production costs are $C(q) = 30,000 + 100q + 0.5q^2$. What is the monopolist's revenue function? What is the monopolist's profit function? What level of output maximizes the monopolist's profits, assuming that the monopolist produces? Should the monopolist produce? What are the monopolist's profits under the optimal decision?

Problem 21 If a monopolist maximized profit per unit, will total profits be maximized? Explain why or why not.

Problem 22 True, False, Uncertain? [Explain your answer.] *The ratio of a monopolist's optimal price to its marginal cost is larger when the market elasticity of demand is greater in absolute value (i.e., more elastic).*

Problem 23 True, False, or Uncertain? [Explain your answer!] *Whenever a firm changes its price in one direction, its revenue changes in the opposite direction if the demand curve is relatively inelastic.*

Problem 24 True, False, or Uncertain? [Explain your answer!] *If the Maximum Pie Company (which is a price-making monopoly) wants to maximize its profit, it should follow this motto: Whenever you can sell an additional pie for more than it costs to make, make more. Whenever you must sell a pie for less than it cost to make it, make less.*

Problem 25 True, False, or Uncertain. [Explain your answer!] When an author sells exclusive rights to a book to a publisher, the publisher becomes a monopoly seller of the book. Assume that, in return for the exclusive rights, the publisher pays the author 10% of the gross revenues from the book (i.e., the author receives $\frac{1}{10}pq$). Assume that demand for the book is neither infinitely elastic nor completely inelastic and that the marginal cost of each additional book is greater than zero. *The author will feel that the publisher charges too much for the book.*

Problem 26 Consider a monopolist with demand given by the relationship $q = 20 - 2p$ and a cost function given by $C(q) = 5q$.

(a). What is the optimal price and quantity for the monopolist?

(b). Illustrate in one figure the MR, MC and demand curves. (You should label the axes and indicate the values of the curves wherever they intersect the axes.) Indicate the deadweight loss from monopoly.

(c). If the monopolist could perfectly 1st-degree price discriminate, how does the monopolist's marginal revenue curve change? What is the new optimal quantity?

Problem 27 The local government decides that it wants cable television in the area. Suppose that the demand curve for cable TV is given by $P(q) = 500 - q$ and that the economic cost for operating a cable company is $C^O(q) = 10,000 + 100q$, where q is the number of households deciding to buy cable TV and p is the price per year for a cable hookup. [Ignore any differences between short-run and long-run costs.]

(a). If the city cares only about increasing city revenues through licenses, should it sell a license to operate cable TV to one firm or several? To achieve this end, should it regulate the price as part of the terms of the license? [Assume that firms will know how many licenses will be sold and whether or not they'll be regulated when they make their decision.]

(b). How much cash can the government raise from the sale of these cable TV rights? [Solve for the dollar amount for full credit.]

(c). Make an economic argument (no need to use math) that if the city knows the cost function of the cable TV industry and if it cares about the sum of its consumers' surplus and the revenues it makes from licensing, that it will wish to license a single cable company with some sort or regulation (e.g. price ceilings) on the price which the cable TV company can charge its customers. In your explanation, indicate why licensing to one firm with price regulation is preferred to licensing to several firms.

Problem 28 Consider a monopolist with opportunity costs given by $C^O(q) = 2q + 2$ and demand given by $q = 20 - 2p$. What is the profit maximizing quantity and price, and corresponding profit?

Problem 29 True, False, or Uncertain. [Explain your answer!] *A monopolist that wishes to maximize profit will always find it optimal to set a price such that all consumers who value the product above marginal cost will purchase the product.*

Problem 30 True, False, or Uncertain? [Explain your answer!] An empirical study of demand elasticities across various industries finds that with monopolies, demand is elastic (i.e., $|\varepsilon| > 1$). *This empirical finding implies that monopolies can only profitably exist in markets in which demand is elastic at any price.*

Problem 31 Suppose that a monopolist's (opportunity) costs are given by the function $C(q) = 10 + 8q$ and the monopolist's demand is given by $q^d = 10 - \frac{1}{2}p$. What is the firm's optimal choice of p and q?; what are the corresponding profits?

Problem 32 True, False, Uncertain? [Explain your answer.] *A profit maximizing monopolist's selling price is always lower than its marginal revenue.*

Problem 33 Suppose that a monopolist faces demand $P(q) = 300 - 6q$ and has marginal cost of $MC(q) = 120 + 6q$.

(a). How many units should this monopolist produce to maximize profits?

(b). What price should this monopolist charge to maximize profits?

(c). What is the dead-weight loss of monopoly in this case?

Problem 34 Suppose that a dominant firm with marginal costs and average costs both equal to 10 faces a competitive fringe of price-taking firms that will supply inelastically exactly 50 units of output. The goods are homogenous, and the market demand function is $Q^d = 200 - 3p$. How much should the large firm supply?

Problem 35 Suppose that a monopolist sells a single good to a population of consumers which is comprised of two types which arise in equal proportion. One type of consumer (type A) values the good at 100; the other type of consumer, type B, values the good at only 75. Each consumer wants at most one unit of the good. Each type of consumer occurs with equal proportion the population. (If it helps, you can let n be the even number of consumers and $n/2$ be the number of each type in the market.)

(a). Assume that the monopolist finds it profitable to operate. If the monopolist's marginal cost of production is 0 and he cannot distinguish between the consumer types, what is the optimal price for the monopolist to charge? (You may assume that a consumer who is indifferent between purchasing and not purchasing will always choose to purchase the good.)

(b). Suppose that the monopolist has a choice between three product designs, all with identical setup costs. Once the product design is chosen, the monopolist sells only that variation of the good. (The monopolist is not designing a product line in this question.) Marginal costs of production are zero under each design. The first product design is the same as before (i.e., type A consumers value the good at \$100 and type B consumers value it at \$75). The second product design appeals to type A consumers: type-A consumers value the good at \$125 while type-B consumers value this design the same as the first product design, at \$75. The third product design appeals more to type-B consumers: type-B consumers value this variant of the good at \$80. Type A consumers, unfortunately, like the third design least of all: they value this version of the good at only \$85. If all these product designs are equally costly to implement and only one can be chosen, and if the marginal costs of production of the good is zero regardless of the design, which product design should the monopolist choose?

(c). Continuing with the facts of part (b): From the point of view of society, which product design generates the most welfare (consumer surplus plus economic profit)? [Hint: the answers to (b) and (c) are different.] Explain this conclusion in contrast to your answer to (b).

Problem 36 Suppose that a dominant firm with marginal costs of $MC(q_m) = 50$ faces a competitive fringe of price-taking firms will to supply according to the supply function $S_f(p) = 150p$. The goods are homogenous, and the market demand curve is

$$p = 600 - \frac{1}{50}Q.$$

What is the monopolist's optimal output?

Problem 37 Suppose that there are two duopolists, "A" and "B", choosing output in a Cournot game. The market demand curve is given by

$$p = 100 - q_a - q_b.$$

Firm A has zero marginal cost, while firm B has a constant marginal cost of $MC_b = 20$. What is the equilibrium price in the market?

Problem 38 Suppose that there are two duopolists, "A" and "B", choosing output in a Cournot game. The market demand curve is given by

$$p = 100 - q_a - q_b.$$

Firm A has zero marginal cost up to a capacity of 15, beyond which it cannot produce. Firm B has a constant marginal cost of $MC_b = 20$ and can produce unlimited amounts at that unit cost. What is the equilibrium price in the market?

Problem 39 Kate and Alice are ready-mix concrete duopolists. The market demand function is

$$Q = 20,000 - 200p,$$

where p is the price of a cubic yard of concrete and Q is the number of cubic yards demanded per year. The marginal cost of producing a cubic yard for each firm is constant at \$80. Kate and Alice compete by choosing quantities and the market price is determined so as to equate their supplies with market demand.

(a) What is Kate's residual demand curve, as a function of Alice's expected supply, $\hat{q}_A$?

(b) What is Kate's optimal quantity as a function of $\hat{q}_A$?

(c) What is the equilibrium market supply of output, $Q^* = q_A^* + q_K^*$?

(d) Suppose now that Kate can publicly commit to a particular output before Alice has a chance to choose her output. What is the (Nash) equilibrium market price in this Leader-Follower quantity game?

Problem 40 Consider a monopolist with demand function $q = 150 - 3p$, and opportunity-cost function, $C^O(q) = 10q$.

(a). What point on the demand curve should the monopolist choose (i.e., what p^* and q^*).

(b). What is the deadweight loss of monopoly in this case?

Problem 41 Consider a dominant firm competing against a competitive fringe of price-taking firms. The supply from the competitive fringe is given by

$$S_f(p) = 19p.$$

The market demand curve is given by

$$p = 1200 - Q,$$

where $Q = q + S_f(p)$. If the dominant firm has constant marginal cost equal to 20, how much should it produce?

Problem 42 Consider two competing duopolists, Tina and Emily. The both sell a homogenous good into a market with demand given by

$$p = 200 - \frac{1}{2}Q.$$

Emily's marginal cost is constant and equal to $MC_e = 15$. Tina has a lower MC cost which is constant at $MC_t = 10$. Tina and Emily play an output game in which they both bring their output to market and the market price is determined from above with $Q = q_e + q_t$ (i.e., this is a Cournot output game).

What is the equilibrium price?

Problem 43 Price discrimination requires the ability to sort customers and the ability to prevent arbitrage. Explain how the following can function as price discrimination schemes and discuss both sorting and arbitrage.

(a). requiring airline travelers spend at least one Saturday night away from home to qualify for a low fare.

(b). selling food processors along with coupons that can be sent to the manufacturer to obtain a \$10 rebate.

(c). offering temporary price cuts on bathroom tissue.

(d). charging high-income patients more than low-income patients for plastic surgery.

Problem 44 (Overheard at a grocery store). "I am getting sick and tired of all these specials, double coupons and the like. If the store would simply discontinue these specials and pass the savings along in lower prices every day, I would be better off." Explain whether you agree or disagree with this statement.

Problem 45 You are an executive for Super Computer, Inc. (SC), which rents out super computers. SC receives a fixed rental payment per time period in exchange for the right to unlimited computing at a rate of p cents per second. SC has two types of potential customers of equal number – ten businesses and ten academic institutions. Business customers have demand functions $q_B = 10 - p$, where q_B is in millions of seconds per month; academic institutions have demands $q_A = 8 - p$. The marginal cost to SC of additional computing is two cents per second, no matter what the volume. Ignore shutdown decisions.

(a). Suppose that you could separate business and academic customers. What rental fee and usage fee would you charge each group? What are your profits?

(b). Suppose you were unable to keep the two types of customers separate, and you charged a zero rental fee. What usage fee (i.e., price) maximizes your profits? What are your profits?

(c). Suppose you set up one two-part tariff; that is, you set one rental and one usage fee that both business and academic customers face. What usage and rental fees will you set? What are your profits? Explain why price is not equal to marginal cost.

Problem 46 Suppose that a firm has two distinguishable customers: students and non-students. The total demand for students is $q_s = 100 - p$. The total demand for non-students is $q_n = 200 - p$. Total costs are $C(q_s + q_n) = \frac{1}{2}(q_s + q_n)^2$. What is the optimal pair of prices which a 3rd degree price-discriminating monopolist would choose? What are profits?

Problem 47 True, False, Uncertain? [Explain your answer.] A Belgian ice cream company decides to expand its sales to Luxembourg. Because the company is less known outside of Belgium, it decides to charge a lower price for its Luxembourg ice cream than for its Belgium ice cream. Assume that it is not profitable for an outside company or individual to arbitrage ice cream across national boundaries. *The price the company charges for its Belgian ice cream should be derived independently of its price for its Luxembourg ice cream.*

Problem 48 Macroware has two types of customers for its computer design software: home users and business users. Each user will purchase only one unit of the software.

Macroware is considering producing two versions of the software. Version A contains the full features of the Macroware software. Version B is the same as Version A in every way, except it does not allow the user to design in color. The willingness to pay of business and home users for Macrowares software is given below:

Version	Business	Home
Version A	\$2000	\$800
Version B	\$1000	\$650

There are 100 business users and 100 home users.

For simplicity, assume also that Macroware incurs zero costs for either version.

(a). If Macroware can tell Business and Home users apart and can charge them different prices, what price will Macroware set for Version A for home users? What price will Macroware set for Version A for business users? Will it also sell version B? If so, at what prices; if not, then explain why not?

(b). Now suppose that Macroware cannot tell business and home users apart and cannot charge them different prices for the same good. However, Macroware can charge different prices for Version A and Version B of the software. How much will Macroware charge for Version A? How much will Macroware charge for Version B?

(c). How might your analysis change in part (b) change if there were 190 business users and 10 home users? (Do not do any calculations, just describe).

Problem 49 Third-degree (or direct) price discrimination tends to be more common in the sale of services (movies, transportation, medical services, etc.) rather than manufactured goods (TV's, stereos, food, etc.). What is a *simple* economic explanation for this?

Problem 50 Suppose that there are two consumers in equal proportion in the marketplace: one which is type A and one which is type B. You are a monopolist, your costs of production are zero, and you cannot distinguish between the two consumer types. The consumer demands are characterized by the following data which lists the maximum prices the various consumers will pay for each additional unit:

Quantity	Consumer A	Consumer B
first	10	12
second	8	10
third	3	7
fourth	2	5

(a). If only a single per-unit price is offered, what is the profit-maximizing price? What is the total profit at this price? [Calculate profits assuming that there is exactly one consumer of each type.]

(b). If a different price can be offered for each level of output, what is the optimal price schedule to offer? (That is, what price will be charged for the first unit, for the second unit, etc.?) What is the total profit? [Calculate profits assuming that there is exactly one consumer of each type.]

Problem 51 Hayley's Bistro serves two types of customers, senior citizens and young people. There are 100 potential customers of each type. The total willingness to pay for dinner at Hayley's by the members of each group is a function of the time of day. Each customer will only dine once per day.

Time	Senior	Young People
Dinner at 6	\$29	\$45
Dinner at 7	\$35	\$58

Assume the costs of serving customers are constant throughout the day and are equal to \$10 per customer. This implies that overcrowding, turning people away, etc. are not problems, even at peak hours.

(a). Suppose that Hayley is forced to charge the same price for dinner at 6 as for dinner at 7. Hayley is also forced to charge the same price to Seniors as to Young People. How much should Hayley charge? How much profit will Hayley earn?

(b). Would your answer in (a) change if it cost \$20 to serve each customer?

(c). Now suppose that Hayley can charge a senior citizen price and a regular price. What are the profit-maximizing prices for senior citizens and young people? How many people total will dine at 6? At 7? What will Hayley's profits be?

(d). Now suppose that Hayley cannot charge different prices to seniors and young people. However, Hayley can charge different prices for dinner at 6 and for dinner at 7. How much should Hayley charge for dinner at 6? For dinner at 7? How much profit will Hayley earn?

(e). Hayley does some more market research for eating at 5pm. She learns that her diners have the following willingness to pay for dinner at various times. (Each diner still only eats at most once per night)

Time	Senior	Young People
Dinner at 5	\$17	\$18
Dinner at 6	\$29	\$45
Dinner at 7	\$35	\$58

Assume that Hayley cannot charge seniors and young people different prices, as in part (d). Also, she thinks that it would be too confusing to have many different prices throughout the evening. Thus, she will choose to offer only the 7pm (regular) time and an early time, each with different prices. Hayley must decide both the prices and what "early" means (5pm or 6pm).

What prices will Hayley charge and which dining time will be required for the lower "early" price? How much profits will Hayley earn?

Problem 52 The producers of consumer goods often issue coupons that let one buy the product at a discount. These coupons are usually distributed as part of an advertisement for the product; they may appear in a newspaper or magazine,or as part of a promotional mailing. For example, a coupon for a particular breakfast cereal might be worth 25 cents toward the purchase of a box of the cereal. We might wonder why firms issue these coupons when printing them and collecting them are expensive. We might wonder why the firm does not just lower the price of the item.

One possibility is price discrimination. Consumers who clip coupons may be more price sensitive than other people. By issuing coupons, a cereal company can separate its customers into two groups, and in effect charge the more price-sensitive customers a lower price than the other customers.

The price elasticities of demand for users vs. nonusers of coupons have been estimated (Narasimhan, "A Price Discrimination Theory of Coupons," *Marketing Science*, 1984). Here are some examples of the price elasticities of demand for users vs. non-users of coupons:

PRICE ELASTICITY	Nonusers	Users
Spaghetti sauces	-1.50	-2.00

Obviously, the elasticity of demand is not constant along the demand curve,but suppose that the elasticity is constant for the purposes of this question. [I've also taken the liberty of rounding Narasimhan's numbers for spaghetti sauce to make computations easy.]

Suppose that a monopoly manufacturer of spaghetti sauce has decided to charge $2 for spaghetti sauce. What face value of coupon should the manufacturer issue? [Assume you are not concerned about non-users of coupons switching to become users of coupons.]

Problem 53 True, False, or Uncertain. [Explain your answer!] Consider a monopolist who price discriminates using third-degree (direct) price discrimination across two markets, A and B. *If at the optimal prices the elasticity of demand is* -2 *in market* A *and* -4 *in market* B, *then the optimal prices will be in a ratio of* $3:2$.

Problem 54 Suppose that you are a monopolist with zero economic costs. You have two types of customers. Some consumers are more price sensitive than others (type A's) and these more elastic consumers also have more time available for collecting coupons for price reductions. The inelastic consumers (type B's) can also collect coupons, but at a higher personal cost. Specifically,

	Consumer A	Consumer B
value of good	10	20
cost of coupon collection	2	5

For example, consumer B will pay 20 for the good without producing a coupon and 15 (i.e., $20 - 5 = 15$) for the good if consumer B has to go through the trouble of collecting the coupon.

(a). Suppose that your customer population consists of equal proportions of each type. What is the optimal price for the good? Do you offer a coupon? If so, how much of a price discount do you offer for the coupon?

(b). Suppose that your customer population consists of twice as many type B consumers as type A consumers. What is the optimal price for the good? Do you offer a coupon? If so, how much of a price discount do you offer for the coupon?

(c). Suppose that the cost of coupon collection is the same for both consumer types. Without doing any calculations, can you say whether or not coupons will be used to maximize profits? Explain.

Problem 55 Mitchell's movie theater has two types of customers: *students* who are not willing to pay much to see a movie at the theater, and regular customers ("*regulars*") who have a higher willingness to pay. There are 100 customers of each type. Mitchel has excess seat capacity (more than 200 seats) and can show movies at both 5pm and 8pm at no additional cost. Both types of customers prefer to watch movies at 8pm rather than 5pm. The willingness to pay for each type of customer at the two possible times is shown below.

	regulars	students
8pm	\$15	\$8
5pm	\$7	\$5

The marginal cost of an additional theater-goer is zero at either showing. *Assume each individual attends at most one movie.*

(a). Suppose Mitchell can tell students and regulars apart. Also suppose that he can set prices based on customer type (student or regular) and time of showing (5pm or 8pm). What prices will he set for each movie type and customer type? Explain. (Recall that a consumer will buy only one movie.) If he decides not to offer a particular time to a particular type of customer, indicate that as well. Explain.

(b). Now suppose that Mitchell cannot tell students and regulars apart. However, Mitchell can charge different prices for the 5pm and 8pm showings. How much will Mitchell charge for 8pm showing? How much will he charge for 5pm showing? Explain.

(c). Suppose that instead of 100 of each type of consumer, there are 150 student customers and only 50 regular customers. How does your answer to (b) change? Explain.

Problem 56 Suppose that a monopolist sells to two markets and can set a different price in each market. The demand functions are

$$q_1 = 200 - 2p_1,$$

$$q_2 = 300 - 2p_2.$$

The monopolist's marginal and average cost is zero up to a capacity of 200 units; the monopolist cannot produce beyond 200 units.

What are the optimal prices to set in each market?

Microeconomics **Professor Lars Stole**
Business 33801

Answers to Week 1 Problems

1 (a). The equilibrium price with 6 licenses will equate demand to supply. Because supply is $\overline{Q} = 60,000$, the equilibrium price is $p^* = 0.60$.

(b). At $p = 0.60$, each firm earns an average profit of $(p - AC) = 0.60 - 0.20 = 0.40$ per mile. Thus, each firm earns a profit of $\pi = 0.40(10,000) = 4000$.

(c). Absent any restriction on entry, price would decrease until $p = AC$, or $p^* = 0.20$. At this price, output would be $0.20 = 1.2 - \frac{Q}{100,000}$ or $Q^* = 100,000$. The deadweight-loss triangle has a base equal to the difference between free-entry supply of 100,000 and the constrained supply of 60,000; thus, the base is 60,000 miles. The height is the difference between the constrained price of $p = 0.60$ and the free-entry price of $p = 0.20$, or 0.40 dollars per mile. All together, we therefore have $DWL = \frac{1}{2}(0.40)(40,000) = 8,000$ dollars.

2 (a). To find the equilibrium, set quantity demanded equal to quantity supplied.

$$500 - 20p = 5p,$$

$$500 = 25p,$$

$$p^* = 20,$$

Hence, $Q^* = 500 - 20p^* = 100$ million DVDs.

(b). In equilibrium, two conditions must hold. First, the market must clear so that $Q^d = Q^S$. Second, there is a wedge of \$5, so $p_s = p_b - 5$. Together, these conditions imply

$$500 - 20(p_s + 5) = 5p_s,$$

$$500 - 20p_s - 100 = 5p_s,$$

$$400 = 25p_s,$$

$$p_s^* = 16.$$

This implies that $p_b^* = p_s^* + 5 = 21$ and the equilibrium quantity will be (substituting into demand) $Q^* = 500 - 20(21) = 80$ millions DVDs.

(c). The story I was looking for was the one discussed in class: trades which produce joint surplus between the buyers and sellers, but which fail to produce enough surplus to cover the \$5 tax (i.e., all trades that generate surplus between 0 and 5) are prevented by the imposition of the tax. The deadweight loss equals the total lost surplus of these trades.

3 False. A decrease in marginal cost is identical to a subsidy on a seller's output. Because the decrease in cost is captured largely by the sellers (they obtain 60% of the decrease, while consumers only obtain 40%), supply must be relatively less elastic compared to demand. Remember that the more elastic side of the market obtains less of the subsidy:

$$\frac{\Delta p_s}{subsidy} = 0.60 = \frac{|\varepsilon_d|}{|\varepsilon_d| + \varepsilon_s}.$$

4 True. A reduction in tax is like an increase in subsidy. Effectively, the government is subsidizing gasoline sales. The subsidy goes proportionately to the more inelastic side of the market. In this case, the demand side gets more of the money. Alternatively, you could have noted that the gasoline tax is paid mainly by the inelastic (consumer) side of the market. A reduction in the tax therefore goes back to that side of the market. (**Aside:** You didn't need to provide the formula, but with these elasticities, the proportion of the reduction that goes to consumers is $1.5/(1.5 + .5) = .75$; consumers get 75 percent of the 5 cents, so almost 4 cents of the reduction.)

5 (a). Before the tax: $100 = 20 + 20p$ so $p = 4$ and $Q = 100,000$.

(b). After the tax, $q = 100,000$ as before, but now $100 = 20 + 20(p - 1)$, and so the after tax price to the buyer is $p = 5$. The seller still receives 4 per carton.

(c). There is no excess burden from this tax because Q is not affected. All transactions conducted before the tax are still conducted after the tax. Deadweight loss occurs when some trades which are socially efficient are not undertaken (or vice versa). Here, it was not sufficient for an ecomomic explanation of *what* is deadweight to say that the demand curve is inelastic.

(d). (i) A positive excess burden will now emerge because some transcations will not take place. Specifically, any consumer who values cigarettes above marginal cost but below marginal cost plus 1 will not purchase. (ii) In addition, the sellers will bear some of the tax now because the buyers are no longer perfectly price inelastic.

6 (a). Before the tax, the equilibrium price is found by equating demand and supply: $100,000 - 40,000p = 20,000$. This is equivalent to $80,000 = 40,000p$. Thus, $p = 2$. Equilibrium quantity is $Q = 20,000$.

(b). The supply and demand elasticities are found by multiplying the relevant dQ/dp by the ratio $p/Q = 1/10,000$. Hence, $\varepsilon^d = -4$ and $\varepsilon^s = 0$ (the latter is true because the supply curve is vertical.

(c). After the tax, supply bears the entire burden of the tax because supply is fixed (perfectly inelastic). Hence, the demand price paid by consumers is still $p_b = 2$, while the supply price received by producers falls to $p_s = 1$, and the equilibrium quantity remains at $Q = 20,000$.

(d). As the result of imposing this tax, the government raises $\$1 \times 20,000 = \$20,000$.

(e). Consumers bear none of the burden of this tax, while producers bear the entire burden of (i.e., all $20,000).

(f). Deadweight loss (a.k.a. excess burden) is zero, since the loss in total (actually, just producer) surplus equals the burden of the tax.

(g). As the supply becomes more elastic, the excess burden increases from zero to a positive number. Both consumers and producers bear a positive fraction of the tax burden, with the precise distribution of the tax burden depending on the demand and supply elasticities. The relatively more inelastic side of this market bears more of the burden.

7 FALSE. In competitive markets, per-unit taxes are not passed along to consumers if either demand is very elastic relative to supply or, stated another way, supply is very inelastic relative to demand. The italicized statement is false or uncertain on at least two counts (you needed to indicate at least one of them): (i) if supply is relatively more elastic than demand, then the result goes the other way; (ii) but it is possible that the italicized statement is uncertain since it is not a relative statement (i.e., even for a very elastic supply curve, if demand is relatively more elastic, consumers will not bear much of the tax).

8 TRUE. Consider the regulated steel market and the unregulated resale market in turn. In the regulated steel market, because the steel companies are paid less for steel, they will produce less steel. In the resale market, the amount of available steel is limited by the amount produced (which is now lower because of the regulation). Graphically, the supply curve for resold steel must shift inward. Because less steel is supplied, the price of steel in the resale market will increase.

Now, consider automobile production. There are two cases. Fist, the auto company may have to go to the secondary steel market to obtain its steel because not everyone will be able to obtain steel at the regulated price. In this case, the cost of steel increases. Providing that marginal cost also increases (a reasonable assumption in the case of automobiles), prices will rise for these automobile manufacturers. Second, consider automobile manufacturers who are lucky enough to obtain the low-priced steel. Here's the subtle part: Even companies that are fortunate enough to get the steel at a low price will still have an opportunity cost of the steel equal to the resale market price, since that is the value of the best alternative use. Hence, the cost of steel as an input into production increases for all automobile manufacturers. This will increase the price of automobiles.

9 (a). Equating demand and supply, we obtain

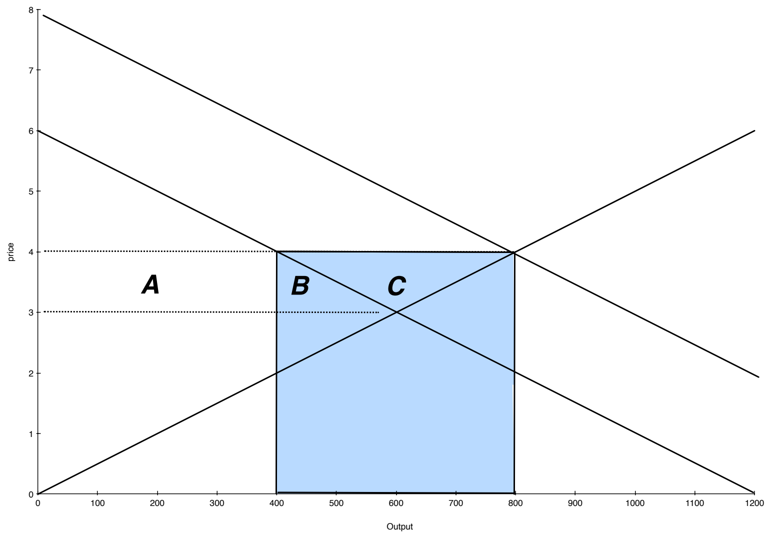
$$1200 - 200p = 200p,$$

so $p^* = 1200/400 = 3$. At this price, the equilibrium quantity is $Q^* = 200(3) = 600$.

(b). In order to raise the market price to $p = 4$, the total demand (consumers plus government) must equal supply at $p = 4$. Thus,

$$Q^d = 1200 - 200(4) + q^d_{govt} = 200(4) = Q^s.$$

Hence, the government must buy $q^d_{govt} = 400$ units and total demand rises to 800 units.



Notice first that the price support reduces consumer surplus by the areas A and B combined. On the other hand, producer surplus is increased by the amount of the areas in A, B and C combined. Thus, total surplus is increased by C. This area is equal to

$$\frac{1}{2}(800 - 400)(4 - 3) = 200.$$

The government price support program, however, costs the amount of the shaded rectangle: $(800 - 400)(4) = 1600$. Hence, $DWL = 1600 - 200 = 1400$. That's the simplest answer to the question.

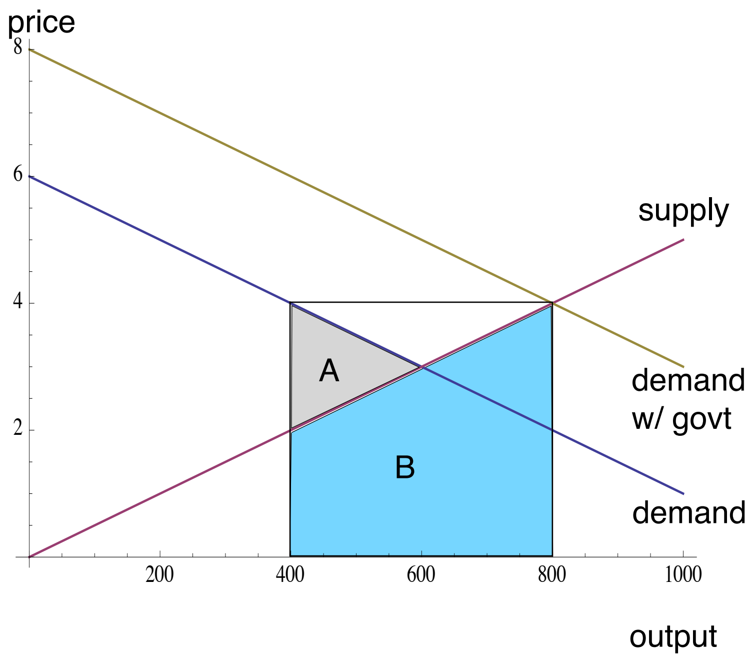
There are a few ways think about the story behind the deadweight loss, and each is equally valid. I think the clearest way to do this is presented in on one of the lecture slides.

Lecture slide approach: You can view the deadweight loss is in two parts. First, let's suppose that the government didn't destroy the coffee but instead gave it to the highest-valuing consumers who did not buy at $p = 4$. The result is that the 400 units of coffee would generate consumer surplus to this group equal to the area under the demand curve, from $Q = 400$ (those who buy at $p = 4$) to $Q = 800$. We thus have a trapezoid with base equal to 200. The left side has height $p = 4$; the right side has height $p = 2$ (because $1200 - 200p = 800$ implies $p = 2$). Thus, the area of the trapezoid is $(1/2)(4 + 2)(400) = 1200$. Thus, if the government gives the coffee away to the highest-value buyers who do not purchase at $p = 4$, the only DWL is the loss from excess trade (this is the triangle to the right of C which is the same type of DWL as arises from unit subsidies):

$$DWL_1 = \frac{1}{2}(4 - 2)(800 - 600) = 200.$$

The fact that the coffee is not given away but instead destroyed, however, generates a second component of deadweight loss, $DWL_2 = 1200$, because the trapezoid of consumer surplus from the government coffee does not arise, so 1200 is wasted.

Second approach: There is another, equally valid, way to decompose the deadweight loss into two components, but these components are different and focus on the cost side. Consider the case in which the government restricted supply to $Q = 400$ and only the most efficient firms are allowed to produce coffee. At this supply, the equilibrium price would be $p = 4$ rather than $p = 3$ and too little coffee would be traded. The deadweight loss of this intervention is equal to the area of triangle "A" in the figure below, which has base 600-400 (the reduction in output) and height $4 - 2 = 2$ (because 400 units are demanded at a price of 4 and 400 units would have been supplied at a price of 2).



The area is $\frac{1}{2}(200)(2) = 200$. Note that the government policy in this question achieves the same outcome $p = 4$ by buying up output. Thus, this 200 will be part of the DWL. But the government in this question has achieved this outcome by purchasing coffee which cost the suppliers to make. The total cost of suppliers to produce the government's 400 units of coffee is the area under the supply curve from $Q = 400$ to $Q = 800$. This is area "B". At $Q = 400$, the height of the supply curve is 2; at $Q = 800$, the height of the supply curve is 4. The area under the supply curve is equal to the area of a trapezoid with these dimensions; thus, the area is $\frac{1}{2}(2 + 4)(800 - 400) = 1200$. Because the government destroys all of this coffee, the cost of producing it is deadweight loss. Hence, the total $DWL = 200 + 1200 = 1400$.

(c). Because the equilibrium post-subsidy prices satisfy $p_s = p_b + 1$ and because $Q^s(p_s) = Q^d(p_b)$, we have the following equilibrium condition:

$$1200 - 200p_b = 200(p_b + 1),$$

and so

$$1200 = 400p_b + 200,$$

or $p_b^* = 2.50$. It follows that $p_s^* = p_b^* + 1 = 3.50$.

10 False. Since the constant-cost industry was in long-run equilibrium initially, each firm was at its minimum long-run average cost. When p is forced below this level, every firm will find it optimal to shut down.

11 False. The subsidy has the effect of shifting the supply curve to the right, since any producer will be willing to produce at least as much as before, and for at least some growers will produce more output given the governmental payments. (This is certainly true in Spain where there are high subsidies for olive production.) When the supply curve shifts right, the market price must decrease. If the demand curve is elastic, then an decrease in the price must increase revenues (i.e., total market expenditures).

A clever answer would add that if the price change moved you from an elastic region of the demand curve to an inelastic region, then it is possible that the statement is true. The question, as written, did not rule out this possibility, but you did not need to mention it to obtain full marks.

12 Standard stuff. See textbook.

13 A complete answer should have noted with a figure or careful verbal argument that (a) does not need to hold, or it should have demonstrated that (b)-(d) must all be true.

Statement (a) is false. Generally, only some of the producer surplus is transferred to consumers. The remaining surplus is destroyed (in deadweight loss) or retained by producers. A figure can easily indicate this logic.

Statement (b) is true because at a binding price ceiling, excess demand will exist (demand will exceed supply). Hence, the market will not clear. Statement (c) is true because at a lower market price, firms who produce earn less revenue per unit (at the same cost) and some firms who used to make profits drop out of the market altogether. In both cases, profits (i.e., producer surplus) are lower. Statement (d) is true because consumers who continue to purchase the good get it for a lower price (and hence consumer surplus is higher), while some consumers can no longer purchase the good, so their consumer surplus is lower. In the extreme case of perfectly inelastic supply, aggregate consumer surplus goes up because all consumers continue to buy. In the other extreme case when the price is set at a level below the minimum AVC for the most efficient supplier, the entire market shuts down and so consumer surplus vanishes (and hence, it is lower in aggregate).

A complete discussion of these issues can be found in the textbook discussion of price ceilings and price floors (GLS, pages 60-68).

14 TRUE. Inelastic goods, by definition, are goods whose transactions change very little when prices (or taxes on prices) are introduced. Since deadweight loss is entirely contained within the interval of lost transactions, inelastic goods are better to tax if the goal is to minimize deadweight loss.

15 TRUE. In a constant cost industry, the long-run supply curve is horizontal at the minimum long-run average cost of each firm. Since a tax has been added to the cost of production, LR average cost must rise by exactly the amount of that tax. Hence, the supply curve shifts upward by 2.

16 Marginal cost is $MC = 4$. Marginal revenue is found by solving for the demand curve as a function of price, $p = 20 - \frac{1}{2}q^d$, and then either (i) applying our linear-demand rule (same 2 intercept and twice the slope) or (ii) multiplying by q and then taking the derivative with respect to q. Either way, we get $MR = 20 - q$. Hence, the optimal q^* is given by $MR(q^*) = MC(q^*)$ or $20 - q^* = 4$, so $q^* = 16$. At this level of output, price is simply $p^* = 20 - \frac{1}{2}q^* = 20 = 8 = 12$. Profit is $\pi = p \cdot q - C(q) = (12)(16) - 16 - 4(16) = 7(16) = 112$.

17 False. The optimal price remains unchanged when demand doubles (although twice as much output is produced). This is easiest to see by using the elasticity version of the marginal-output rule. The elasticity of demand for the monopolist is initially

$$\varepsilon^d = \frac{dq^d}{dp}\frac{p}{q} = D'(p)\frac{p}{D(p)}.$$

Once demand doubles, the new elasticity of demand is

$$\varepsilon^d = \frac{dq^d}{dp}\frac{p}{q} = 2D'(p)\frac{p}{2D(p)} = D'(p)\frac{p}{D(p)}.$$

Hence, the elasticity of demand (as a function of price) does not change when the market demand doubles. Hence,

$$\frac{p^* - MC}{p^*} = \frac{1}{\varepsilon^d}$$

doesn't change because MC is constant. Thus, the optimal price also doesn't change. (Of course, the optimal output increases from $D(p^*)$ to $2D(p^*)$, so output doubles.)

18 False. A firm will choose output to equate marginal cost and marginal revenue. Unless this happen by coincidence to be the same output which minimizes long-run average cost, average cost will not be at its minimum.

19 (a). The goods are demand substitutes because the quantity demanded of one good as a function of the price of the other good is positively related.

(b). Let's solve this problem both ways. Maximizing over quantities involves more steps, but is closer to the simple monopoly model from class. Maximizing over prices is more direct, but we will no longer have an explicit expression for marginal revenue with respect to output).

- **Maximizing over outputs:** Notice that the marginal revenue of coffee for the firm depends upon the amount of coffee sold *and* the amount of espresso sold. This arises because the demands of the goods are not independent. Indeed, the goods are demand substitutes in this question. An increase in the sales of coffee, for example, requires a lower price of coffee, which in turn will reduce demand for espresso. To be clear, let's write out the revenue contribution from coffee:

$$p_c \cdot q_c = \left(6 - \frac{3}{8}q_c - \frac{1}{8}q_e\right) q_c = \frac{1}{8}q_c(48 - 3q_c - q_e).$$

Similarly, the revenue from espresso sales is

$$p_e \cdot q_e = \left(6 - \frac{1}{8}q_c - \frac{3}{8}q_e\right) q_e = \frac{1}{8}q_e(48 - q_c - 3q_e).$$

All together then, revenue (as a function of outputs) is

$$R(q_c, q_e) = \frac{1}{8}\left(q_c(48 - 3q_c - q_e) + q_e(48 - q_c - 3q_e)\right).$$

The marginal revenue of coffee, therefore, requires that we differentiate with respect to q_c:

$$MR_c(q_c, q_e) = \frac{1}{8}\left(48 - 6q_c - q_e\right) - \frac{1}{8}q_e = \frac{1}{8}\left(48 - 6q_c - 2q_e\right).$$

Notice that the additional q_e term captures the marginal revenue effect of q_c on the revenues from espresso. A similar computation gives us the marginal revenue of espresso:

$$MR_e(q_c, q_e) = \frac{1}{8}\left(48 - 2q_c - 6q_e\right).$$

Equating these marginal revenues to their respective marginal costs, we have two marginal output equations:

$$\frac{1}{8}\left(48 - 6q_c - 2q_e\right) = 0.50,$$

$$\frac{1}{8}\left(48 - 2q_c - 6q_e\right) = 1.50.$$

Solving these two equations for q_c^* and q_e^* yields $q_c^* = 6$ and $q_e^* = 4$. With these quantities, the corresponding prices are

$$p_c^* = \left(6 - \frac{3}{8}(6) - \frac{1}{8}(4)\right) = 3.25,$$

$$p_e^* = \left(6 - \frac{1}{8}(6) - \frac{3}{8}(4)\right) = 3.75.$$

- **Maximizing over prices:** Maximizing over prices is much more direct in this setting. Let's write the firm's payoff as a function of prices:

$$\begin{aligned}
\pi(p_c, p_e) &= D_c(p_c, p_e)(p_c - 0.50) + D_e(p_c, p_c)(p_e - 1.50) \\
&= \left(12 - 3p_c + p_e\right)\left(p_c - 0.50\right) + \left(12 + p_c - 3p_e\right)\left(p_e - 1.50\right).
\end{aligned}$$

Differentiating this expression with respect to the price of coffee, p_c, yields the marginal profit (with respect to price) of

$$\begin{aligned}\frac{\partial \pi(p_c, p_e)}{\partial p_c} &= D_c(p_c, p_c) + \frac{\partial D_c(p_c, p_e)}{\partial p_c}(p_c - 0.50) + \frac{\partial D_e(p_c, p_e)}{\partial p_c}(p_e - 1.50) \\ &= (12 - 3p_c + p_e) - 3(p_c - 0.50) + (p_e - 1.50) \\ &= 12 - 6p_c + 2p_e.\end{aligned}$$

To maximize profits, prices should be set so that this margin is zero. Hence, we require that

$$12 - 6p_c + 2p_e = 0.$$

A similar (but equally tedious) calculation for the marginal effect of p_e establishes that

$$16 + 2p_c - 6p_e = 0.$$

Solving these two price equations simultaneously, we obtain $p_c^* = 3.25$ and $p_e^* = 3.75$, just as before.

20 The revenue function is given by

$$R(q) = pq = (1000 - q)q = 1000q - q^2.$$

The monopolist's profit function is

$$\pi(q) = R(q) - C(q) = 1000q - q^2 - 30,000 - 100q - 0.5q^2 = 900q - 1.5q^2 - 30,000.$$

The optimal output level, assuming the $q^* > 0$ is optimal, is found at $\pi'(q^*) = 0$. That is,

$$\pi'(q^*) = 900 - 3q^* = 0.$$

The solution to this equation is $q^* = 300$. Alternatively, the optimal output can be found by setting $MR(q) = 1000 - 2q$ equal to $MC(q) = 100 + q$. Thus,

$$1000 - 2q = 100 + q,$$

so $q^* = 300$. To determine whether the monopolist should produce at all, we can check that $AR = p = 1000 - 300 = 700$ is greater than $AC = \frac{30,000}{300} + 100 + 0.5(300) = 350$. Thus, $q^* = 300$ is optimal. Alternatively, we can check the actual value of profits at $q^* = 300$.

$$\pi(q^*) = (1000 - 300)(300) - 30,000 - 100(300) - 0.5(300)(300) = 105,000.$$

21 Maximizing profit per unit does not maximize total profits. Maximizing profit per unit is equivalent to maximizing $P(q) - AC(q) = P(q) - C(q)/q$. The output that maximizes profit per unit is generally different from the output that maximizes profit, $qP(q) - C(q)$. The former requires setting $MR(q) - MC(q) = P(q) - AC(q)$ which will not equal zero except in the unique situation where the profit-maximizing output earns exactly zero profits.

22 FALSE. It decreases. This is immediate for a perfect competitor (i.e., $\varepsilon = -\infty$) in which case the price equals marginal cost. More generally, the relationship is

$$\frac{P(q)}{C'(q)} = \frac{\varepsilon}{\varepsilon + 1},$$

which increases as ε becomes closer to unit elasticity, $\varepsilon = -1$; i.e., as the market becomes less elastic. NB: a monopolist always operates where $\varepsilon \leq -1$.

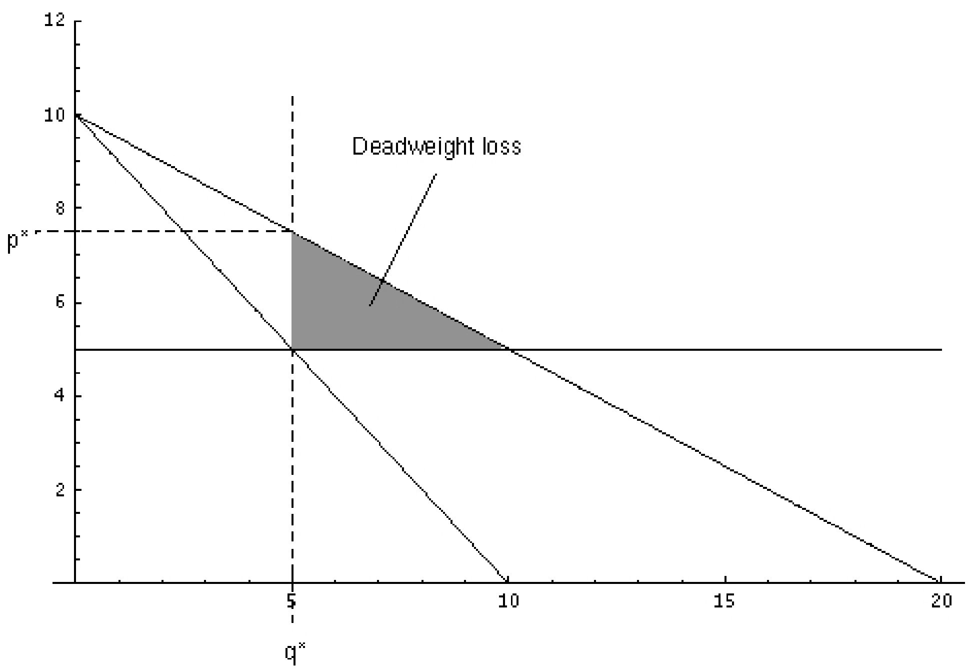
23 FALSE. When demand is inelastic, a price increase raises revenue and a price decrease reduces revenue. Thus, revenue and prices move together.

24 FALSE. This motto would require that you produce where $P(q) = MC(q)$. But a monopolist should set production to the point where $MR(q) = MC(q) < P(q)$.

25 True. Because the author does not bear any fraction of the marginal cost of production, she will want to maximize revenues. Since a monopolist will set a price where demand is elastic, the price will be too high from the point of view of revenue maximization.

26 (a). Note that $P(q) = 10 - \frac{1}{2}q$, so $MR = 10 - q$. $MC = 5$. Hence, the optimal quantity is found by setting $10 - q = 5$, so $q^* = 5$. The corresponding optimal price is $p^* = 10 - \frac{1}{2}q^* = 7.50$.

(b).



(c). With perfect first-degree price discrimination, the monopolist prices at each buyer's marginal valuation, hence, $MR(q) = P(q) = 10 - \frac{1}{2}q$. The optimal quantity is found by equating this to maginal cost: $10 - \frac{1}{2}q = 5$, or $q^* = 10$.

27 (a) Potential cable TV firms are only willing to pay as much as they expect to earn in profits from the license to operate a cable TV service. The highest level of expected profits can be obtained by a monopolist charging the monopoly price. Thus, the city should sell to a single firm an unregulated license.

(b) The most cash which the city can raise is equal to the profits of the monopolist. Here, the monopolist's profits are calculated in the usual way: first, find the optimal price, and second, find out what profits are at that optimal price. Profits (excluding the license fee) are given by

$$\pi = (500 - q)q - 10,000 - 100q.$$

The first-order condition is (MR=MC)

$$500 - 2q - 100 = 0.$$

Thus, the monopolist would choose to sell to only $q = 200$ households at a price of $p = 500 - 200 = \$300$ per year. Profits at this price and quantity are

$$\pi = (300)(200) - 10,000 - 100(200) = \$30,000.$$

Thus, the city can charge a license fee of $30,000.

(c) If the city cares about the firm's profits (which it does indirectly through its influence on the license fee it can charge) and about consumer surplus, the town will want to regulate price. Indeed, if the town cares only about the sum of the two (i.e., social surplus), it will prefer to regulate the price at $p = 100$ and subsidize the cable company on the order of $10,000 to provide the cable services. Competition (selling to two or more firms) would also improve consumer surplus, but at a social cost of $10,000 per firm in terms of fixed costs.

28 Note that to determine marginal revenue, we must get p on the left-hand side: $p = 10 - \frac{1}{2}q$. Now, as always for linear demand, MR is given by $MR = 10 - q$ (i.e., same intercept but twice the slope). Hence, because MC=2, we have $2 = 10 - q$ or $q^* = 8$ and therefore $p^* = 6$. Profit is simply revenue less costs: $\pi = (6)(8) - (2)(8) - 2 = 30$.

29 FALSE. The monopolist would like to lower the price so as to sell to some of the consumers who value the good above marginal cost but below the monopolist's current price. Here there are clear gains from trade. But to lower the price to serve these consumers requires that the monopolist also lower the price for the higher valuation consumers, and so the monopolist would lose money from the inframarginal consumers. It is the simple monopolist's inability to separate consumers that leads the monopolist to set a price above marginal cost and introduce a deadweight loss into the market. Only a perfect price discriminating monopolist will find it optimal to lower price until every consumer who values the good above marginal cost actually consumes the good.

30 False. A monopolist that follows the marginal output rule will always price in the elastic region if marginal costs are even slightly positive. There were many minor correct things which you could also add to this statement, but given that we discussed the point in class, I required that you state that the monopolist prices in the elastic region to get full credit.

31 Marginal cost is $MC = 8$. Marginal revenue is found by solving for the demand curve as a function of price, $p = 20 - 2q$, and then either (i) applying our linear-demand rule (same intercept and twice the slope) or (ii) multiplying by q and then taking the derivative with respect to q. Either way, we get

$$MR = 20 - 4q.$$

Hence, the optimal q^* is given by $MR(q^*) = MC(q^*)$ or $20 - 4q^* = 8$ or $q^* = 3$. At this level of output, price is simply $p^* = 20 - 2q^* = 20 - 2(3) = 14$. Profit is $\pi = p^*q^* - C(q^*) = (14)(3) - 10 - 8(3) = 8$. [Since the profit is positive, our monopolist will not shutdown. As this was a monopoly problem, you did not need to indicate this for full credit, but you did need to get $\pi = 8$.]

32 False. A monopolist's marginal revenue is given by

$$MR(q) = P(q) + qP'(q).$$

Because the second component is negative when $q > 0$ and demand slopes downward, we know that

$$MR(q) < P(q).$$

Hence, marginal revenue is always less than the selling price.

Intuitively, selling an extra unit generates the price of that unit *less* the loss in revenue on inframarginal units; therefore, marginal revenue is always less than price.

33 (a). Choose q^* to satisfy $MR(q^*) = MC(q^*)$. In our present setting, $MR(q) = 300 - 12q$. Hence, we choose q^* to solve

$$300 - 12q^* = 120 + 6q^*.$$

Solving for q^*, we find that $q^* = 10$.

(b). At an output of $q^* = 60$, the optimal price is found by substituting $q^* = 10$ into the demand function:

$$p^* = P(q^*) = P(10) = 300 - 6(10) = 240.$$

(c). Under marginal-cost pricing, the socially efficient output is obtained:

$$P(q) = 300 - 6q = MC(q) = 120 + 6q,$$

$$180 = 12q,$$

so $q^{fb} = 15$. Because a monopolist only produces $q^* = 10$, we know the dead-weight loss lies between $q = 10$ and $q = 15$ in the standard textbook graph of monopoly deadweight loss; it is bounded above by the demand curve and below by the marginal cost curve. We need only to know the height of this triangle to calculate its area. The top point at $q = 10$ is $p = 240$, as found in (b). The bottom point at $q = 10$ is the marginal cost at $q = 10$: $MC(10) = 120 + 5(10) = 180$. Hence, we have a triangle that is $240 - 180 = 60$ dollars/unit high and 5 units wide. Its area is

$$DWL = \frac{1}{2}(60)(5) = 150.$$

34 First, we compute the large-firm's residual demand function:

$$q^d(p) = (200 - 3p) - 50 = 150 - 3p.$$

With this demand function, we then compute the optimal output and price. The firm's residual demand curve is $p = 50 - \frac{1}{3}q$. Using the marginal output rule , we have

$$MC = 10 = MR(q) = 50 - \frac{2}{3}q,$$

so $q^* = 60$. The associated price is $p^* = 50 - \frac{1}{3}(60) = 30$.

35 (a). It is optimal to choose $p^* = 75$ and sell to every consumer. If the monopolist sells at $p = 100$, he sells to only half of the market and makes an average per-consumer profit of \$50. If the monopolist sells at $p = 75$, he will sell to everyone and make a per-consumer profit of \$75.

(b). The original design generates a per-consumer profit of $\pi/n = \$75$. If the second design is chosen, the monopolist would still choose a price of $p^* = 75$, and thus monopoly profits would be unchanged. The monopolist is indifferent between these two options. The third design, however, would let the monopolist charge a higher price, $p^* = 80$, which would raise per-consumer profits to $\pi/n = 80$. Thus, the product design appealing most to the type B consumers is chosen.

(c). Society is certainly better off choosing the type-A preferred design (design 2) over the original design since the monopolist's profits are unchanged but type-A consumers get more surplus. Design 2 is also socially better than design 3 since the latter would generate only slightly more per-consumer profits $\Delta\pi/n = 5$, but a much lower consumer surplus of $\Delta CS/n = (50 - 5)/n = 45/n$.

The explanation for this contrast with (b) is that the monopolist will choose quality to appeal to the marginal consumer (since this consumer determines the price, and hence profits). In our setting, this is the type B consumer. Social surplus, however, is maximized by considering all of the consumer and producer surplus (not just the impact on the marginal transaction); thus, a full consideration of type A and type B's preferences come into play.

36 We begin by constructing the monopolist's residual demand curve.

$$p = 600 - \frac{1}{50}\left(S_f(p) + q_m\right) = 600 - \frac{1}{50}(150p + q_m) = 600 - 3p - \frac{1}{50}q_m.$$

Simplifying, we have

$$p = 150 - \frac{1}{200}q_m.$$

Marginal revenue is

$$MR_m(q_m) = 150 - \frac{1}{100}q_m.$$

Setting $MR = MC$, we have

$$150 - \frac{1}{100}q_m = 50.$$

Thus,

$$100 = \frac{1}{100}q_m,$$

or $q_m^* = 10,000$. Aside: Note that at this output, the market price will be $p = 150 - \frac{1}{200}(10000) = 100$. In addition, the competitive segment will produce $S_{comp}(100) = 15,000$. Total output is $Q = 10,000 + 15,000 = 25,000$.

37 Firm A chooses q_a to maximize $q_a(100 - q_a - \hat{q}_b)$, given its belief that firm B will produce $\hat{q}_b$. Firm B chooses q_b to maximize $q_b(100 - \hat{q}_a - q_b) - 20q_b$, given its belief that firm A will produce $\hat{q}_a$. Thus, we have two marginal-output equations that must be satisfied in the Nash equilibrium to this game:

$$100 - 2q_a - q_b = 0,$$

$$100 - q_a - 2q_b = 20.$$

Solving these two equations in two unknowns yields $q_a^* = 40$ and $q_b^* = 20$, thus $p^* = 100 - 40 - 20 = 40$.

38 Notice that the setting is exactly as in the previous question except that firm A is constrained to choose $q_a^* \leq 15$. Given the unconstrained solution is $q_a^* = 40$, let's guess that $q_a^* = 15$ in fact arises, and check for B's optimal choice. Firm B, thinking that A will be constrained and choose $q_a^* = 15$, maximizes $q_b(100 - 15 - q_b) - 20q_b$. Marginal revenue for firm B is

$$MR_b(q_b) = (100 - q_a^*) - 2q_b = 85 - 2q_b^*,$$

which is set equal to marginal cost,

$$85 - 2q_b = 20,$$

so $q_b^* = 32.5$. At this output, we can check that firm A's marginal revenue at $q_a^* = 15$ is

$$MR_a(q_a) = (100 - q_b) - 2q_a = (100 - 32.5) - 2(15) = 37.5 > MC_a = 0,$$

so firm A would indeed like to increase its output, but is constrained to choose $q_a^* = 15$. These outputs, therefore, are the Nash equilibrium to the Cournot game. At these outputs, the market price is $p^* = 100 - 15 - 32.5 = 52.50$.

39 (a). $p = 100 - \frac{\hat{q}_A}{200} - \frac{q_K}{200}$. The market inverse demand curve is

$$p = 100 - \frac{1}{200}(q_A + q_K).$$

If Alice is expected to bring $\hat{q}_A$ yards to the market, then Kate's residual demand curve is

$$p = 100 - \frac{1}{200}(\hat{q}_A + q_K) = 100 - \frac{\hat{q}_A}{200} - \frac{q_K}{200}.$$

(b). $q_K^* = 2000 - \hat{q}_A/2$. Because Kate's residual demand curve is

$$p = 100 - \frac{1}{200}(\hat{q}_A + q_K) = 100 - \frac{\hat{q}_A}{200} - \frac{q_K}{200},$$

her (residual) marginal revenue curve is

$$p = 100 - \frac{\hat{q}_A}{200} - \frac{q_K}{100}.$$

Setting this equal to marginal cost, we have the equation for Kate's optimal output (as a function of Alice's expected supply, $\hat{q}_A$:

$$80 = 100 - \frac{\hat{q}_A}{200} - \frac{q_K}{100}.$$

Thus, the best response function for Kate is

$$R_K(q_A) = 2000 - \frac{\hat{q}_A}{2}.$$

(c). $Q^* = 8000/3$. Kate's best response function is

$$R_K(\hat{q}_A) = 2000 - \frac{\hat{q}_A}{2}.$$

Because Kate and Alice have symmetric costs in this problem, an identical argument establishes that Alice's best-response (as a function of her expectation of Kate's supply, $\hat{q}_K$) is

$$R_A(\hat{q}_K) = 2000 - \frac{\hat{q}_K}{2}.$$

A Nash equilibrium requires that these optimal choices are mutually consistent $q_A^* = R_A(q_K^*)$ and $q_K^* = F_K(q_A^*)$, and thus we solve two equations for two unknowns and obtain

$$q_A^* = q_K^* = \frac{4000}{3},$$

Total output is $Q^* = \frac{8000}{3}$.

(d). $p = 85$. Recall that Alice's best-response function is $R_A(\hat{q}_K) = 2000 - \frac{\hat{q}_K}{2}$. Kate, knowing this, can compute her residual demand curve:

$$q_K + R_A(q_K) = 20,000 - 200p \implies q_K + (2000 - q_K/2) = 20,000 - 200p,$$

so

$$p = 90 - \frac{q_K}{400}.$$

Kate's (residual) marginal revenue function is

$$90 - \frac{q_K}{200},$$

and so the marginal output rule implies $80 = 90 - \frac{q_K}{200}$ or $q_K^* = 2000$. Given this supply, Alice will respond with $q_A^* = R_A(2000) = 2000 - \frac{2000}{2} = 1000$. Total output will be $Q^* = 3000$ and the corresponding price will be $p^* = 100 - \frac{3000}{200} = 85$.

40 (a). We need the marginal revenue function to answer this question. First, rearrange the demand function to have price on the left hand side: $p = 50 - \frac{1}{3}q$. Using our result from class,

$$MR(q) = 50 - \frac{2}{3}q.$$

Sett this equal to MC implies

$$50 - \frac{2}{3}q = 10.$$

Thus,

$$40 = \frac{2}{3}q,$$

or $q^* = 60$. At this quantity, the corresponding price on the demand curve is $p^* = 50 - \frac{1}{3}(60) = 30$.

(b). Deadweight loss is the area between the demand curve and the marginal cost curve, between the monopoly output, q^*, and the efficient output, q^{eff} (the output for which the MC and demand curves intersect. This is a triangle. The height of the triangle is $(p^* - MC) = 30 - 10 = 20$. To find the length of the base, note that MC crosses the demand curve at the quantity for which

$$10 = 50 - \frac{1}{3}q,$$

so $q^{eff} = 120$. Because the base length is $(q^{eff} - q^*) = 120 - 60 = 60$, the area of the DWL triangle is

$$DWL = \frac{1}{2}(30 - 10)(120 - 60) = 600.$$

41 First we determine the residual demand curve for the dominant firm.

$$p = 1200 - (q + S_f(p)) = 1200 - q_s - 19p.$$

Solving for p on the lefthand side,

$$20p = 1200 - q,$$

or

$$p = 60 - \frac{1}{20}q.$$

Now we apply the standard monopoly recipe. The residual MR for the dominant firm is

$$MR = 60 - \frac{1}{10}q.$$

Setting $MR = MC$ yields

$$60 - \frac{1}{10}q = 20,$$

or $q^* = 400$.

42 Let's first consider Emily's problem. If she thinks Tina is bringing $\hat{q}_t$ to market, then her residual demand curve is

$$p = 200 - \frac{1}{2}(q_e + \hat{q}_t) = \left(200 - \frac{1}{2}\hat{q}_t\right) - \frac{1}{2}q_e.$$

It follows that Emily's marginal revenue function is

$$MR_e = 200 - \frac{1}{2}\hat{q}_t - q_e.$$

Setting this equal to her marginal cost, the marginal output rule is

$$200 - \frac{1}{2}\hat{q}_t - q_e = 15.$$

Solving for q_e, we have Emily's best response function:

$$q_e = BR_e(\hat{q}_t) = 185 - \frac{1}{2}\hat{q}_t.$$

Now consider Tina's problem. If she thinks that Emily is bringing $\hat{q}_e$ to market, then Tina's residual demand curve is

$$p = 200 - \frac{1}{2}(\hat{q}_e + q_t) = \left(200 - \frac{1}{2}\hat{q}_e\right) - \frac{1}{2}q_t.$$

It follows that Tina's marginal revenue function is

$$MR_t = 200 - \frac{1}{2}\hat{q}_e - q_t.$$

Setting this equal to her marginal cost, $MC_t = 10$, the marginal output rule is

$$200 - \frac{1}{2}\hat{q}_t - q_e = 10.$$

Solving for q_e, we have Tina's best response function:

$$q_t = BR_t(\hat{q}_e) = 190 - \frac{1}{2}\hat{q}_e.$$

The final step is to solve for (q_e^*, q_t^*) that are jointly optimal against one another. Formally, we want to find (q_e^*, q_t^*) that satisfies

$$q_e^* = 185 - \frac{1}{2}q_t^*,$$

$$q_t^* = 190 - \frac{1}{2}q_e^*.$$

We therefore need to solve these two equations for the two unknown values. The algebra is straightforward (though tedious). One can check that $q_e^* = 120$ and $q_t^* = 130$ is the solution. With these outputs, $Q^* = 250$, and thus $p^* = 75$.

43 (a). The requirement of staying over a Saturday night separates business travelers (who prefer to return home for the weekend) from tourists (who travel on the weekend). Arbitrage is not possible when the ticket specifies the name of the traveler.

(b). Rebate coupons with food processors separate consumers into two groups: (1) customers who are less price sensitive (those who have a lower elasticity of demand) do not request the rebate; and (2) customers who are more price sensitive (those who have a higher demand elasticity) request the rebate. The latter group could buy the food processors, send in the rebate coupons, and resell the processors at a price just below the retail price without the rebate. To prevent this type of arbitrage, sellers could limit the number of rebates per household.

(c). A temporary price cut on bathroom tissue is a form of intertemporal price discrimination. Price-sensitive consumers buy greater quantities of tissue than they would otherwise during the price cut. Non-price-sensitive consumers buy the same amount of tissue that they would buy without the price cut. Arbitrage is possible, but the profits on reselling bathroom tissue cannot compensate the cost of storage, transportation, and resale.

(d). The plastic surgeon might not know who is a high-income patient and who is a low-income patient, but can guess. One strategy is to quote a high price initially, observe the patient's reaction, and then negotiate the final price. (Many medical insurance policies do not cover elective plastic surgery.) Because plastic surgery cannot be transferred from low-income patients to high-income patients, arbitrage is no problem.

44 Disagree. This is a form of price discrimination charging a high price to persons with a high value of time and a low price to persons with a low value of time. It is not clear that prices would fall.

45 (a). Set a two-part tariff for each group. For academic customers, consumer surplus with marginal cost pricing is $0.5(8-2)(6,000,000) = 18$ million cents. Therefore, charge \$180,000 per month in rental fees and two cents per second in usage fees, i.e., marginal cost. For business customers, consumer surplus is $0.5(10-2)(8,000,000) = 32$ million cents. Therefore, charge \$320,000 per month in rental fees and two cents per second in usage fees. Total profits for 10 customers of each are $10(320,000+180,000) = \$5,000,000$ per month.

(b). Total demand for the two types of customers with 10 customers per type is $Q = 10(10-p)+10(8-p) = 180-20p$. Solving for price as a function of quantity: $p = 9-Q/20$. Marginal revenue is $9-Q/10$. To maximize revenue, set marginal revenue equal to marginal cost, i.e., $9-Q/10 = 2$ and solve for quantity: $Q = 70$. At this quantity, price is 5.5 cents per second. Profit is 70 million seconds times $(5.5-2)$, or \$2.45 million per month.

(c). With a two-part tariff and no price discrimination, set the rental fee (F) to be equal to the consumer surplus of the academics: $F = 0.5(8-p)(8-p) = 0.5(8-p)^2$. Total revenue when both groups are served is equal to $20(F) + 10(q_A + q_B)p$. Total cost is $20(q_A + q_B)$. Substituting for quantities in the profit equation with total quantity in the demand equation (above in (b)) yields

$$\pi(p) = 10(8-p)^2 + (p-2)(180-20p).$$

Differentiating with respect to p yields

$$\frac{d\pi(p)}{dp} = -20(8-p) - 20(p-2) + 180 - 20p = 0,$$

or $p^* = 3$ cents per second. At this price the rental fee is $0.5(8-3)^2 = 12.5$ million cents or \$125,000 per month. At this price, $q_A = 8-3 = 5$ millions seconds and $q_B = (10-3) = 7$ millions seconds. Profits are $20(12.5) + 10(12)(3-2) = 370$ million cents or \$3.7 million. Since this is greater than in the case where you set a two-part tariff so high that only the business customer participates ($10(32) = 320 < 370$ from part (a)), charging a two-part tariff which induces both groups to participate is optimal. Here, price is not equal to marginal cost because all of the consumer surplus cannot be captured with a two-part tariff.

46 We consider setting two prices (or two quantities), one for each customer segment. Profits are given by

$$\pi(q_s, q_n) = q_s(100-q_s) + q_n(200-q_n) - \frac{1}{2}(q_s+q_n)^2.$$

Differentiating with respect to q_s and q_n yields

$$100 - 3q_s - q_n = 0,$$

$$200 - 3q_n - q_s = 0.$$

Solving these equations simultaneously results in $q_s = 12.5$ and $q_n = 62.5$. The corresponding prices are $p_s = 87.5$ and $p_n = 137.50$. Profits are $(12.5)(87.5) + (62.5)(137.5) - 0.5(75)^2 = 6875.00$.

47 UNCERTAIN. Assuming the firm is engaging in price discrimination, the firm wants to equate the marginal revenue in each market to marginal cost. If the expansion into the Luxembourg market changes marginal cost (e.g., there is only a fixed capacity for production), then the Belgian price should also be changed. If and only if marginal costs are constant across markets can prices be derived independently.

48 (a). For the business users, Macroware will charge $p = 2000$ and sell 100 copies of version A to the segment; for the home users, Macroware will charge $p = 800$ for version A and sell 100 copies to this segment. Macroware will not sell version B because it generates less surplus and Macroware has the ability to extract all the surplus.

(b). Let's rearrange the willingness-to-pay data as a sequence of consumption from the basic good (version B) and an upgrade (which converts the basic good into version A).

Version	Business	Home
Basic good (B)	\$1000	\$650
Upgrade	\$1000	\$150

To maximize revenue on the basic good, it can either set a price of $p_b = 650$ and sell to both types (making revenue of $200 \cdot 650 = 130,000$), or it can set a price of $p_b = 1000$ and sell to only the business customers (making revenue of $100 \cdot 1000 = 100,000$). It is optimal, therefore, to set $p_b^* = 650$. For the upgrade, it can choose to sell it at $p_{up} = 150$ and make $200 \cdot 150 = 30,000$ or to sell it at $p_{up} = 1000$ and make $100,000$. It is therefore optimal to set the upgrade price to $p_{up}^* = 1000$. Hence, it is optimal for Macroware to sell version B for $p_b^* = 650$ and version A for $p_a^* = p_b^* + p_{up}^* = 650 + 1000 = 1650$.

(c). Following the logic above, if there were 190 business users and only 10 home users, it would make sense to sell only to the business users. For example, you would obtain revenue of 190(2000)=380,000 rather than the revenue of 190(1650)+10(650)=320,000 using the prices from (b).

49 Services are difficult to transfer between consumers (i.e., arbitrage), while manufactured goods are easily traded. Also correct, it is easier to tell customer types in face-to-face transactions, which are more typically service oriented.

50 (a). At a price of $p = 12$, only type B buys, so $\pi = 1(12) = 12$. At $p = 10$, type A buys 1 unit and type B buys 2 units: $\pi = 3(10) = 30$. At $p = 8$, type A buys 2 units and type B buys 2 units: $\pi = 4(8) = 32$. At $p = 7$, type A buys 2 units and type B buys 3 units: $\pi = 5(7) = 35$. At $p = 5$, type A buys 2 units and type B buys 4 units: $\pi = 6(5) = 30$. At $p = 3$, type A buys 3 units and type B buys 4 units: $\pi = 7(3) = 21$. At $p = 2$, type A buys 4 units and type B buys 4 units: $\pi = 8(2) = 16$. Hence, the optimal single price is $p = 7$ and profit is $\pi = 35$.

(b). As in class (and in Dolan and Simon), we find the optimal price for each additional unit. For the first unit, the optimal price is $p = 10$ at which 2 units are sold yielding profits on the first unit of 20 (as opposed to choosing a price of $p = 12$ at which only one unit would be sold yielding only 12). Hence, $p_1 = 10$. Similarly, we find that $p_2 = 8$ (with second-unit profit of 16 rather than 10 for a price of 10), $p_3 = 7$ (with profit of 7 rather than 6 for a price of 3), and $p_4 = 5$ (with profit of 5 rather than 4 at a price of 2). Total profit with optimal nonlinear pricing is therefore $\pi = 20 + 16 + 7 + 5 = 48$.

51 (a). Two candidates present themselves as possible optimal prices. First, if the price was set to 35, 200 customers would eat at the restaurant. Profits would be (35-10)200=5000. Alternatively, setting a price at 58, only 100 customers would show up; profit would be (58-10)100=4800. Hence, a price of 35 is optimal and profits are 5000.

(b). When costs are 20 rather than 10, the answer would differ from part (a). The profit associated with a price of 35 is now only (35-20)200=3000, while the profit associated with a price of 58 is $(58 - 20)100 = 3800 > 3000$. Hence, the price should be raised to 58.

(c). This is a case of direct (3rd-degree) price discrimination. Given you can price by customer type, you will want to serve both senior citizens and young customers at their preferred times (7pm), but charge them different prices: $p_s = 35$, $p_y = 58$. Profits are

$$\pi = 100(58 - 10) + 100(35 - 10) = 7300.$$

(d). The correct answer is $p_6 = 29$ and $p_7 = 42$. At these prices, the seniors will to dine at 6pm, and the young diners will dine at 7pm. Profits are

$$\pi = 100(42 - 10) + 100(29 - 10) = 5100.$$

The most straightforward manner to arrive at these optimal prices is to convert the problem into one that looks like our airline pricing example from class. The "basic" (low-quality) good is dining at 6pm, while the upgrade is converting the basic dining experience into the 7pm dinner. With this reorganization of the goods, we have

	Senior	Young People
Basic good (6pm)	\$29	\$45
Upgrade	\$6=(35-29)	\$13=(58-45)

The basic good should be priced at either 29 or 45. At a price of 29, all 200 diners will consume it and profit will be $200(29 - 10) = 3800$. At a price of 45, only 100 diners will choose it, and so $\pi = 100(45 - 10) = 3500 < 3800$. Hence, $p = 29$ is optimal.

In terms of the upgrade, we perform a similar analysis. At a price of 6, 200 diners will choose it and dine at 7pm generating profit $200(6) = 1200$; at a price of 13, however, profit will be greater at $100(13) = 1300 > 1200$. Thus, the "upgrade" is priced at 13. Recombining our prices into our original goods, we have $p_6 = 29$ and $p_7 = 29 + 13 = 42$.

(e). We now have an expanded set of options compared to part (d). Obviously, we could still choose early and regular prices of 29 and 42 from part (d) by choosing the early-bird time to be any dinner starting at 6pm or earlier, and obtain the same profit as in (d) which is 100(29-10)+100(42-10)=1900+3200=5100.

There new possibility to consider is that Hayley could set an early bird time of 5pm. We will compute the optimal prices in this new setting and the associated profit using the same technique as in (d). Then we can pick the setting which delivers the highest profit.

The relevant table with a 5pm early dining option is now

	Senior	Young People
Basic good (5pm)	\$17	\$18
Upgrade	\$18=(35-17)	\$40=(58-18)

For the early dinner, a price of $p = 17$ will generate profit of $200(17 - 10) = 1400$; a price of $p = 18$ will generate profit of $100(18 - 10) = 800$ which is lower, and so $p_5 = 17$ is optimal. For the upgrade price, a price of $p_u = 18$ will generate profits of $200(18) = 3600$; a price of $p_u = 40$ will generate profit of $100(40) = 4000$ which is higher, so $p_u = 40$ is optimal, and thus the optimal prices are $p_5^* = 17$ and $p_7^* = 17 + 40 = 57$. With these prices, Hayley makes profit of $\pi = 100(17-10)+100(57-10) = 5400$. This is higher than 5100 in (d), so we know that an early-bird time of 5pm is better than 6pm.

52 The marginal revenue for a monopolist equals $p(q)\left[1 - \frac{1}{\epsilon}\right]$. Because this is set equal to marginal cost, and the marginal cost of a jar of sauce is the same regardless of the purchaser, we expect

$$p_u(q)\left[1 - \frac{1}{\varepsilon_u}\right] = MC = p_n(q)\left[1 - \frac{1}{\varepsilon_n}\right].$$

Plugging in the values of elasticities, we have

$$p_u(q)\left[1 - \frac{1}{2}\right] = p_n(q)\left[1 - \frac{1}{1.5}\right].$$

Simplifying (and using $p_n = \$2$),

$$p_u\frac{1}{2} = 2\frac{1}{3}.$$

Thus, $p_u = \frac{4}{3}$. This price requires that users have a coupon for 67 cents off (or 33% off, in percentage terms).

53 True. Marginal cost will be equated across the two markets, so

$$p_A\left(1 + \frac{1}{\varepsilon_A}\right) = p_B\left(1 + \frac{1}{\varepsilon_B}\right),$$

or more simply $p_A(1/2) = p_B(3/4)$, so $p_A/p_B = 3/2$.

54 (a). There are a variety of ways to solve this problem. We will follow the approach of Dolan and Simon, discussed in class. First, we need to reorganize the preference information in terms of units of consumption. The first unit is the good with coupon collection requirement; the second item is the additional value of removing the coupon collection requirement. Hence, we have

Units	**Consumer A**	**Consumer B**
net value of good with coupon collection	8	15
additional value of removing coupon requirement	2	5

We need to find two prices, p_1 and p_2, where p_1 is the price with coupon redemption and p_2 is the additional price paid without coupon redemption. First, consider p_1. With equal proportions, a price of $p_1 = 8$ generates an average revenue of 8 per consumer as every consumer purchases the basic good. A price of $p_1 = 15$ generates only $\frac{1}{2}(15) = 7.5$, as only the type B consumers (half the population) purchase the basic good. Hence, $p_1^* = 8$. Now consider the second "unit". Here, a price of $p_2 = 2$ generates a revenue of 2 per consumer, while a price of $p_2 = 5$ generates a revenue of $\frac{1}{2}(5) = 2.5$ per consumer. Hence, $p_2^* = 5$ is optimal. The result is that the retail price of the good should be $p = p_1 + p_2 = 13$ and the coupon discount should be for 5. The type A consumers collect coupons, pay $13 - 5 = 8$ and get no consumer surplus; type B consumers do not collect coupons and pay the retail price of 13.

(b). Repeating the previous analysis, our answer changes when we consider $p_1 = 15$. Now a $\frac{2}{3}$ fraction of the population is of type B, so a price of $p_1 = 15$ generates an expected revenue of $\frac{2}{3}(15) = 10$ per consumer, which exceeds 8. Thus, $p_1^* = 15$. As before, we also find that $p_2^* = 5$. Thus, the solution involves a price of $p = 20$ for the good and no coupons are redeemed in equilibrium. In equilibrium, type B consumers alone consume the good, and they do not waste their time collecting coupons. (Hence, the coupon value is less than or equal to 5).

(c). If both types suffer the same discomfort from coupon collection, then coupons cannot be used as a screening device. The coupon collection requirement cannot be used as an implicit test to distinguish among the consumer types because all consumers are impacted the same way. Furthermore, because coupons only waste trading surplus and lower the effective price that consumers will pay, there is no reason to require coupon redemption.

55 (a). It is profit maximizing to only show the 8pm showing to charge a price of $p_s = 8$ to students and a price of $p_r = 15$ to regular movie goers. The 5pm showing does not take place because both types prefer 8pm and there is excess capacity at 8pm.

(b). We can answer this by viewing the product in two components. The basic movie product (which equals the 5pm film) and an upgrade to the preferred time of 8pm. Viewed in this way, we have the following table of values:

	regulars	students
5pm	\$7	\$5
upgrade	\$8	\$3

Now we can apply our methodology from class and price the components row by row. The 5pm film is optimally priced at $p = 5$ which generates revenue of $5(200) = 1000$ which is greater than pricing at $p = 7$ and selling to only 100 customers. The preferred-time upgrade is optimally priced at $p_u = 8$ and sold to only 100 customers for revenue $8(100) = 800$ rather than priced at 3 and sold to 200 customers for the lower revenue of $3(200) = 600$. Translating these prices back into the prices for the 5pm and 8pm showings, we have $p_5^* = 5$ (the basic good) and $p_8^* = p_5 + p_u = 5 + 8 = 13$.

(c). Using the same table as in (b), we recompute the row-prices. For the basic good, $p = 5$ as before. For the upgrade, however, it is now optimal to price at $p_u = 3$ which yields revenue $3(200) = 600$ rather than a price of $p_u = 8$ which would generate lower revenue equal to $8(50) = 400$. Hence Mitchell wants to offer only the 8pm showing to both types of movie goers for price $p_8^* = 8$. Because we want all consumers to go to the 8pm showing, Mitchell should show only one movie at 8pm as in part (a).

56 Let's first solve for prices on the lefthand side in each market:

$$p_1 = 100 - \frac{1}{2}q_1,$$

$$p_2 = 150 - \frac{1}{2}q_2.$$

If ignoring the capacity constraint, the firm finds it optimal to produce levels which satisfy the constraint, then we have a simple direct price discrimination model with $MC = 0$. If you do this exercise, you will find that $q_1 = 100$ and $q_2 = 150$, which violates the firms capacity constraint, so this cannot be the solution.

Thus, what we know is that $q_1^* + q_2^* = 200$, and we will want to set the marginal revenues equal across the segments:

$$100 - q_1 = 150 - q_2,$$

or using $q_2 = 200 - q_1$, we have

$$100 - q_1 = 150 - (200 - q_1),$$

and thus $q_1^* = 75$. This implies that $q_2^* = 125$ (because $q_1 + q_2 = 200$). The corresponding optimal prices are

$$p_1^* = 100 - \frac{1}{2}(75) = 62.50,$$

$$p_2^* = 150 - \frac{1}{2}(125) = 87.50.$$